

COMP 532

Machine Learning and Bioinspired Optimisation

Lecture 1



UNIVERSITY OF
LIVERPOOL

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Welcome to COMP532

- Overview of the module
- Introduction to bio-inspired learning
- Applications

Overview

- Module delivery
 - Lecturer: Meng Fang (Meng.Fang@liverpool.ac.uk)
 - Demonstrators
- Module Website
 - <https://liverpool.instructure.com/courses/85682>
 - Module github: <https://github.com/mengfn/COMP532-Machine-Learning-and-Bioinspired-Optimisation>
 - All lecture materials are available on Canvas:
 - Slides
 - Assignments
 - Tutorials and Coding Solutions
 - Papers and Notes

Overview

- Delivery of Lectures
 - 30 one-hour lectures in total
 - 3 lectures each week (on campus)
- Delivery of Tutorials
 - One-hour each week (on campus)
 - Tools
 - Python
 - Google Colab: <https://colab.research.google.com/>

Overview

- Lectures will be recorded and uploaded to Canvas each week
 - stream.liv.ac.uk
 - Canvas module - COMP532
- MS Teams
- Meeting hours
 - Friday, 1:00 pm – 2:00 pm

Module Aims and Objectives

- We focus on **learning agents** that **interact** with an initially unknown world. Since the world is **dynamic** this module will put strong emphasis on learning to deal with sequential data unlike many other machine learning courses.

Module Aims and Objectives

- Introduce and give an overview to the state-of-the-art **bio-inspired self-adapting** methods.
- Enable students to not only learn to build models with reactive input/output mappings but also **build computer programs** that sense and perceive their environment, plan, and make optimal decisions.
- Familiarise students with **reinforcement learning, multi-agent learning, swarm intelligence, deep learning, evolutionary game theory**.
- Demonstrate **principles of bio-inspired methods**, provide indicative examples, develop problem-solving abilities and provide students with experience to apply the learnt methods in real-world problems.

Learning Outcomes

- A systematic understanding of bio-inspired algorithms
- In depth insight in the mathematics of biologically inspired machine learning and optimisation methods
- A comprehensive understanding of the benefits and drawbacks of the various methods
- Demonstrate knowledge of using the methods in real-world applications
- Practical assignments will lead to hands on experience

Module Topics

- Problem solving from Nature
- Reinforcement learning
- Deep learning
- Multi-agent systems

Module Syllabus (approximate)

- Problem Solving from Nature (2)
- Math and Programming Basics (1)
- Reinforcement Learning (7)
- Deep Learning (7)
- Revision (1)
- Multi-Agent System (7)
- Swarm Intelligence (4)
- Revision (1)

Module Assessments

- Written exam: 70%
- Two course projects: Each 15%
 - Will be available in Canvas – Module COMP532 Assignments
 - Pen & Paper, Programming tasks
 - Groups of 3 (or more)

Assessment Projects

- Task 1 on reinforcement learning, some programming and pen/paper exercises (15%)
- Task 2 on deep reinforcement learning implementation and pen/paper exercises (15%)

Useful Resources

- Reinforcement learning

- Courses

- UCL Course on RL, <https://www.davidsilver.uk/teaching/>
 - Reinforcement Learning: A Graduate Course, https://www.it.uu.se/research/systems_and_control/education/2017/relearn

- Book

- Reinforcement Learning: An Introduction, by Richard S. Sutton and Andrew G. Barto, <https://web.stanford.edu/class/psych209/Readings/SuttonBartoIPRLBook2ndEd.pdf>

- Machine learning

- Stanford, CS229 Machine Learning, <https://cs229.stanford.edu/>

- Deep learning

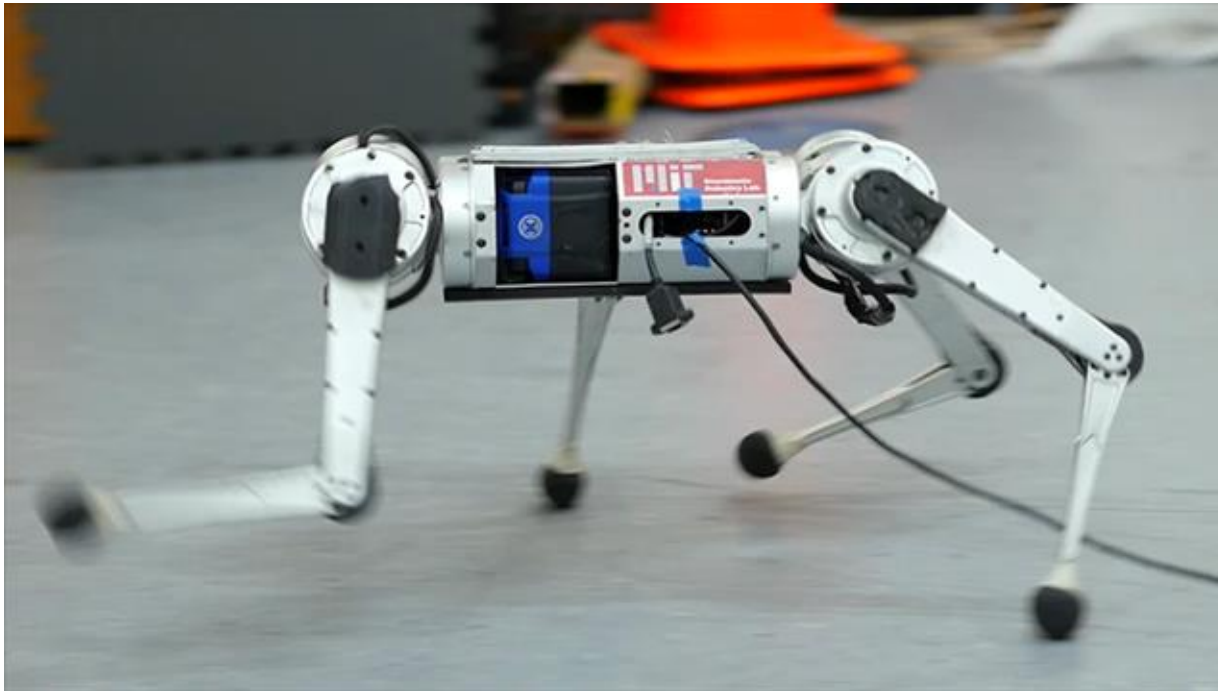
- MIT, Introduction to Deep Learning, <http://introtodeeplearning.com/>
 - Stanford, CS230 Deep Learning, <https://cs230.stanford.edu/>
 - Book

- Deep Learning, by Ian Goodfellow and Yoshua Bengio and Aaron Courville, <https://www.deeplearningbook.org/>

Introduction to Bio-inspired Learning

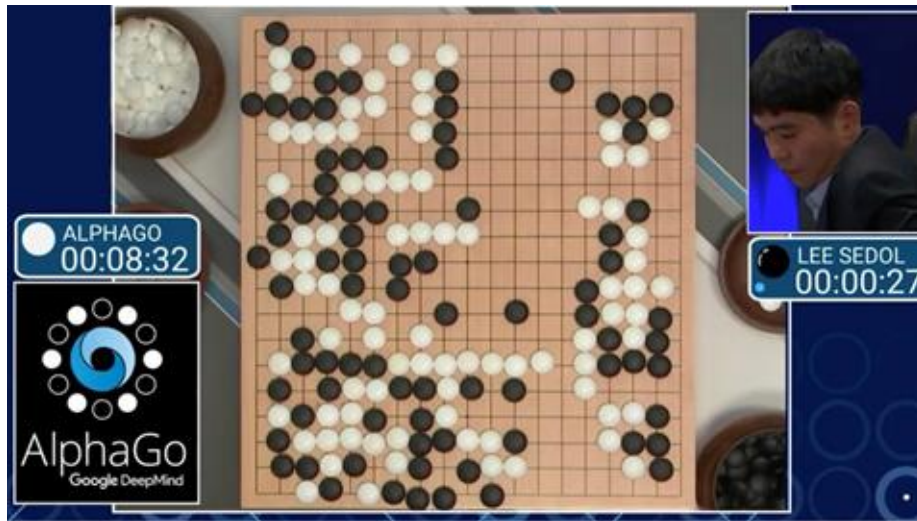
- Bioinspired Robotics

- [Bioinspired Robotics: Smarter, Softer, Safer](#)
- Bioinspired robots walk, swim, slither and fly, <https://www.nature.com/articles/d41586-022-03014-x>



Nature or Bio-inspired Learning

- How a robot/computer/agent makes decisions?



Nature or Bio-inspired Learning

- Bio-inspired computing
 - Solve computer science problems using models of biology
 - In computer science, bio-inspired computing relates to artificial intelligence and machine learning
- Examples:
 - Reinforcement Learning
 - Multi-agent learning
 - Evolutionary Game Theory
 - Evolutionary Computation – Swarm Intelligence
 - ...
- What is Agent?

Nature or Bio-inspired Learning

- Agent
 - An independent program or entity that interacts with its environment by perceiving its surroundings via sensors, then acting through actuators or effectors.



Nature or Bio-inspired Learning

- This course is NOT about Advanced Machine Learning or Data Mining for its Applications
- But it is about **agents** situated in a (potentially) unknown **environment**, learning **how to behave and optimize** its action in order to achieve its goals, taking **inspirations from nature**.

Nature or Bio-inspired Learning

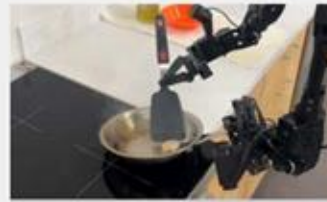
- Some important notes:
 - We are not necessarily mimicking nature
 - Inspirational

Nature or Bio-inspired Learning

- Single agent reinforcement learning
 - Application example: [Mobile ALOHA: Your Housekeeping Robot](#)



Learned Policies



cook shrimp



push chairs



use cabinet



wipe wine



call elevator



high five

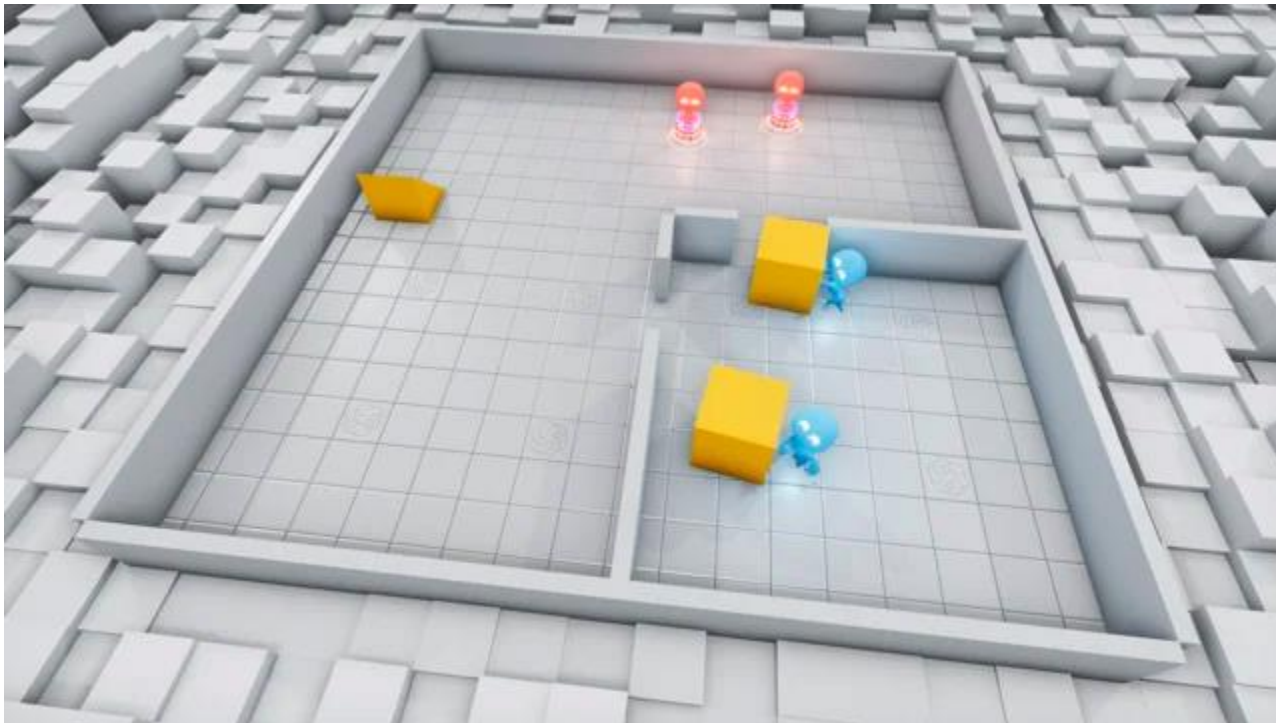
Nature or Bio-inspired Learning

- Deep reinforcement learning
 - Application example: [DQN Breakout](#)



Nature or Bio-inspired Learning

- Multi-agent learning
 - Application example: [Multi-Agent Hide and Seek](#)



Nature or Bio-inspired Learning

- Swarm intelligence
 - Application example: [Teaching groups of drones to fly in formation](#)



Nature or Bio-inspired Learning

- Other applications
 - Portfolio optimisation
 - Playing games better than humans
 - Search engines
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Summary

- Introduction to the course
- Topics of the course
- Introductory applications/examples